

MANTHAN

NEET 2027

PHYSICAL CHEMISTRY

CHEMICAL KINETICS

LECTURE -03

Subtopic - order of reaction

- Sunday → liquid solution
- 25 Question test
- Qsd-sheet
- NEET-PYQ
- NEET Adv. Books
- JEE-MAINS DPP
- test

Sarvesh Sir

Today's goal



► ORDER OF REACTION

► MOLECULARITY



→ Test → Sunday = 2 pm = 15 Aug.
telegram पर है।

Sarveshdixidcsdofficial.

→ Answer Key = 8 pm (evening)



Calculation of order of Reaction. If some data is given.

1. Unit of Rate Constant



Rate law Expression

$$R.O.R. = \text{Rate} = k [A_t]^\eta \rightarrow \text{order of rxn}$$

$$k = \frac{\text{Rate}}{[A_t]^\eta} = \frac{(\Delta C / \Delta t)}{[A_t]^\eta} = \frac{\left(\frac{\text{mole/litre}}{\text{Sec}} \right)}{\left(\frac{\text{mole}}{\text{litre}} \right)^\eta} = \frac{(\text{mole/litre})^1}{\left(\frac{\text{mole}}{\text{litre}} \right)^\eta \cdot \text{Sec}}$$

$$k = \left(\frac{\text{mole}}{\text{litre}} \right)^{1-\eta} \cdot \text{Sec}^{-1}$$

$$K = [\text{Con.}]^{1-\eta} \cdot \text{Sec}^{-1}$$

$$\frac{x^1}{x^\eta} = x^{1-\eta}$$

Csd →

$$\begin{aligned}\text{unit of } k &= [\text{Con}]^{1-\eta} \cdot \text{Sec}^{-1} \\ &= \left[\frac{\text{mole}}{\text{litre}} \right]^{1-\eta} \cdot \text{Sec}^{-1}\end{aligned}$$

$$\begin{aligned}\text{Rate} &= \frac{[\text{Con}]}{\text{time}} = \frac{\text{mole/litre}}{\text{Sec}} \\ &\text{mole} \cdot \text{litre}^{-1} \cdot \text{Sec}^{-1}\end{aligned}$$

Where $\rightarrow \eta = \text{order of Reaction}$

$$\eta = 0 \rightarrow \text{zero order Rxn} \rightarrow k = \left(\frac{\text{mole}}{\text{litre}} \right)^{1-0} \cdot \text{Sec}^{-1} = \boxed{\text{mole litre}^{-1} \text{Sec}^{-1}}$$

$$\eta = 1 \rightarrow \text{1}^{\text{st}} \text{ order Rxn} \rightarrow k = \left(\frac{\text{mole}}{\text{litre}} \right)^{1-1} \cdot \text{Sec}^{-1} = \boxed{\text{Sec}^{-1}}$$

$$\eta = 2 \rightarrow \text{2}^{\text{nd}} \text{ order Rxn} \rightarrow k = \left(\frac{\text{mole}}{\text{litre}} \right)^{1-2} \cdot \text{Sec}^{-1} = \left(\frac{\text{mole}}{\text{litre}} \right)^{-1} \cdot \text{Sec}^{-1} = \boxed{\text{mole}^{-1} \text{litre} \cdot \text{Sec}^{-1}}$$

Identify the reaction order from each of the following rate constants. Example 4.4

(i) $k = 2.3 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1} \rightarrow 2^{\text{nd}} \text{ order rxn}$

(ii) $k = 3 \times 10^{-4} \text{ s}^{-1}$

(i) The unit of second order rate constant is $\text{L mol}^{-1} \text{ s}^{-1}$, therefore $k = 2.3 \times 10^{-5} \text{ L mol}^{-1} \text{ s}^{-1}$ represents a second order reaction.

Solution

(ii) The unit of a first order rate constant is s^{-1} therefore $k = 3 \times 10^{-4} \text{ s}^{-1}$ represents a first order reaction.

Solⁿ 1 $K = \left(\frac{\text{mole}}{\text{litre}} \right)^{1-\eta} \cdot \text{Sec}^{-1} \quad \left(\frac{x}{y} \right)^{-1}$
 $k = \left(\frac{\text{mole}}{\text{litre}} \right)^{1-2} \cdot \text{Sec}^{-1} \Rightarrow \eta = 2 \rightarrow \left(\frac{\text{mole}}{\text{litre}} \right)^{-1} \cdot \text{Sec}^{-1} = \frac{x^{-1}}{y^{-1}}$

$$= \frac{\text{mole}^{-1}}{\text{litre}^{-1}} \cdot \text{Sec}^{-1} = \underline{\text{mole}^{-1} \cdot \text{litre} \cdot \text{Sec}^{-1}}$$

$$\eta = 1 \rightarrow K = \text{Sec}^{-1}$$

If Relation b/w Con. of Reactant & Rate of rxn is Given



$$R_{\text{old}} = K [A_t]^\eta$$

If Con. of Reactant change = $\underline{x}$ time $\rightarrow x \times A_t$

$$R_N = K [\underline{x} A_t]^\eta$$

$$R_N = K \cdot x^\eta \cdot [A_t]^\eta$$

$$R_N = x^\eta \cdot \underline{K [A_t]^\eta}$$

$$R_N = \underline{x}^\eta \cdot R_{\text{old}}$$

Con. of Reactant change ' $\underline{x}$ ' time
than Rate of rxn change

$\underline{x^\eta \cdot \text{time}}$

$\underline{x=2} \rightarrow \text{double} \rightarrow R_N = 2^\eta R_{\text{old}}$

$\eta=0 \rightarrow R_N = R_{\text{old}}$

$$R_N = x^0 \cdot R_{\text{old}}$$
$$R_N = R_{\text{old}}$$

$$\eta = 1 \rightarrow R_N = 2^1 \cdot R_{old}$$

$$R_N = \textcircled{2} \cdot R_{old}$$

$$\rightarrow R_N = x^1 \cdot R_{old}$$

$$R_N = x \times R_{old}$$

$$\eta = \underline{2} \rightarrow R_N = x^2 R_{old}$$

$$R_N = \{x^2\} R_{old}$$

$$x = 2 \rightarrow \text{double}$$

$$R_N = (2)^2 R_{old}$$

$$R_N = \underline{4} \times R_{old}$$

$$\eta = \underline{3} \rightarrow R_N = x^3 \cdot R_{old}$$

Csd $\rightarrow$ Con. को x time करते हैं
 $R_N = x^n \cdot R_{old}$ change होगा

$$\eta = 0 \rightarrow R_N = R_{old}$$

$$\eta = 1 \rightarrow R_N = x \times R_{old} \quad \left(\text{जो Con. में change वही Rate में change} \right)$$

$$\eta = 2 \rightarrow R_N = x^2 R_{old}$$

$$\eta = 3 \rightarrow R_N = x^3 R_{old}$$



If con. of A is double than
Rate of rxn is increase 2-times
Calculate order of rxn

~~a) 1~~

$$R = k[A_t]^n$$

b) 2

$$R = k[A_t]^2$$

c) $\frac{1}{2}$

$$R = k[A_t]$$

d) $\frac{3}{2}$



data given

S.N.	A_t	Rate
1	0.4M	1×10^{-5}
2	0.8M	1×10^{-5}
3	0.2M	1×10^{-5}

Diagram showing relationships between rows:
 - From row 1 to row 2: $\times 2$ (concentration), $\rightarrow$ (rate)
 - From row 1 to row 3: $\frac{1}{2}$ (concentration), $\rightarrow$ (rate)

Write Correct Rate law Expression

Soln $R = k[A_t]^0$

$$\boxed{R = k}$$

22. For the elementary reaction, $M \rightarrow N$, the rate of disappearance of M increases by a factor of 8 upon doubling the concentration of M. The order of the reaction with respect to M is [2014 Adv.]

(a) 4

(c) 2

~~(b) 3~~

(d) 1

Solution:

$$R_N = x^n R_{old}$$

$$R_N = (2)^n R_{old} \quad (2)^3 = 8$$

$$= 8 R_{old}$$

23. For the non-stoichiometric reaction, $2A + B \rightarrow C + D$, the following kinetic data were obtained in three separate experiments, all at 298 K

[2014, Main]

$$1+0=1$$

	Initial concentration <u>A</u>	Initial concentration <u>B</u>	Initial rate of formation of C (mol L ⁻¹ s ⁻¹)
I. ✓	0.1 M	0.1 M	1.2×10^{-3}
II	0.1 M	0.2 M	1.2×10^{-3}
III ✓	0.2 M	0.1 M	2.4×10^{-3}

(a) $\frac{dC}{dt} = k[A][B]$

(b) $\frac{dC}{dt} = k[A]^2[B]$

(c) $\frac{dC}{dt} = k[A][B]^2$

(d) $\frac{dC}{dt} = k[A]$

$$R = k[A]^x[B]^y$$

$$R = k[A]^1[B]^0$$

$$R = k[A]^1$$

Solution:



$$R = k[A_t]^x [B_t]^y$$

S.N-1 $\rightarrow \underline{1.2 \times 10^3} = k(0.1)^x (0.1)^y$ — (i)

$\rightarrow 1.2 \times 10^3 = k(0.1)^x (0.2)^y$ — (ii)

$\rightarrow 2.4 \times 10^3 = k(0.2)^x (0.1)^y$ — (iii)

$$\frac{(i)}{(ii)} \rightarrow \frac{1}{1} = \frac{(0.1)^y}{(0.2)^y}$$

$$\frac{1}{1} = \left(\frac{0.1}{0.2} \right)^y$$

$$\frac{1}{1} = \left(\frac{1}{2} \right)^y$$

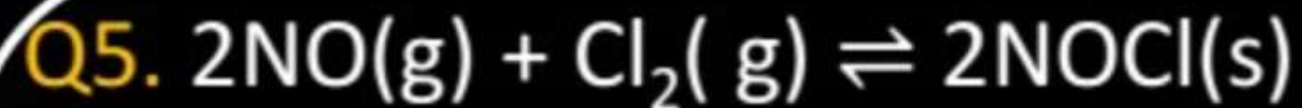
$$\left(\frac{1}{2} \right)^0 = \left(\frac{1}{2} \right)^y$$

$$\boxed{y=0}$$

$$\frac{(i)}{(iii)} \rightarrow \frac{1}{2} = \frac{(0.1)^x}{(0.2)^x}$$

$$\left(\frac{1}{2} \right)^1 = \left(\frac{1}{2} \right)^x$$

$$\boxed{x=1}$$



This reaction was studied at -10°C and the following data was obtained

run	$[\text{NO}]_0$	$[\text{Cl}_2]_0$	r_0
1	0.10	0.10	0.18
2	0.10	0.20	0.35
3	0.20	0.20	1.40

$[\text{NO}]_0$ and $[\text{Cl}_2]_0$ are the initial concentrations and r_0 is the initial reaction rate. The overall order of the

reaction is $1+2=3$

(Round off to the Nearest Integer).

[18 March (Shift 1) 2021]

Q. The result given in the below table were obtained during kinetics studies of the following reaction
 $2A + B \rightarrow C + D$

X and Y in the given table are respectively

IITJEE -2022-23

Experiment	[A] / ^① molL ⁻¹	[B] / ^② molL ⁻¹	Initial rate / molL ⁻¹ min ⁻¹
^① I	0.1	0.1	6.00×10^{-3}
II	^{2x} 0.1	^{x2} 0.2	2.40×10^{-2} ^{x4}
^③ III	0.2	0.1	1.20×10^{-2}
IV	X	0.2	7.20×10^{-2}
V	0.3	Y	2.88×10^{-1}

a) 0.4, 0.3

~~b) 0.3, 0.4~~

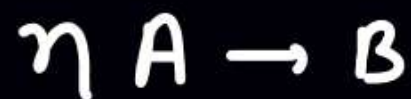
c) 0.4, 0.4

~~d) 0.3, 0.3~~

$$0.2 \times 2 = \underline{0.4}$$

$y = 0.4$
 $x = 0.3$

Effect of volume of solⁿ. in Rate of rxn.



$$\underline{R} = k [A_t]^n \rightarrow \text{order of Rxn}$$

↓
Con. of Reactant

$$[A_t] = \frac{\text{Mole of Reactant}}{\text{Volume of sol}^n. (L)}$$

$$\downarrow [Con] \propto \frac{1}{\uparrow \text{Vol. of sol}^n}$$

$$\text{Rate} \propto [Con]^n \propto \left(\frac{1}{\text{Vol. of sol}^n} \right)^n$$

$$V = 2 \text{ (double)} \rightarrow C = \frac{1}{2}$$

$$V = \frac{1}{2} \text{ (half)} \rightarrow C = 2 \text{ (double)}$$

Q

The rate of a gaseous reaction is given by the expression $k[A][B]$, if the volume of the reaction vessel is suddenly reduced to $1/4^{\text{th}}$ of the initial volume, the reaction rate relating to original rate will be:

A $1/10$

B $1/8$

C 8

~~D 16~~

$$R_0 = k[A][B]$$

$$R_N = k[4A][4B]$$

$$R_N = 4 \times 4 \times \underline{k[A][B]}$$

$$\underline{R_N = 16 \times R_0}$$

$$\uparrow \text{Con.} \propto \frac{1}{\text{vol.}} \downarrow$$

$$C = \frac{1}{V}$$

$$C_N = \frac{1}{\frac{V}{4}}$$

$$C_N = 4 \times \left(\frac{1}{V}\right)$$

$$C_N = 4 \times C$$



CHEMICAL KINETICS

Qsd-SHEET-3

<u>Q-1</u>	$[A_t]$	$[B_t]$	Rate (mole·litre ⁻¹ Sec ⁻¹)
	0.3 M	0.4 M	1×10^2
	0.3 M	0.8 M	2×10^2
	0.9 M	0.4 M	9×10^2

Calculate Rate law expression of Reaction $2A + 2B \rightarrow A_2B_2$

a) $R = k[A_t]$

b) $R = k[A_t]^1[B_t]^1$

c) $R = k[A_t]^2[B_t]$

d) $R = k[A_t]^2$



Rate = $k[A]^2$. If suddenly volume of container increases 4 times, what is the effect of Rate of Reaction

a) No Effect

b) decreases 4 times

c) increases 16 times

d) decreases 16 times

Q-3 Rate law expression of above Reaction



What is Change in Rate of Reaction, when Con. of A is double and Con. of B is half.

- a) Remains same
- b) increases 2 times
- c) decreases 50%.
- d) decreases $\frac{3}{2}$ times

Q-4



[P]

[Q]

Rate (mole·litre⁻¹·sec⁻¹)

0.1M

0.1M

1×10^3

0.2M

0.1M

2×10^3

0.1M

0.2M

4×10^3

0.2M

0.2M

R

find out value of 'R'.

a) 1×10^3

b) 2×10^3

c) 4×10^3

d) 8×10^3

Q-5



[P]	[Q]	Rate (mole·litre ⁻¹ ·sec ⁻¹)
0.1M	0.1M	1×10^3
0.2M	0.1M	2×10^3
0.1M	0.2M	4×10^3
0.2M	x	8×10^3

find out value of 'x'

a) 0.1M b) 0.2M c) 0.4M d) 0.3M

CHEMICAL KINETICS

Qsd-SHEET-2

Q-1 2 $A \rightarrow B$ is an example of elementary reaction

☒ a) Molecularity = 2

☒ b) order of rxn = 2

☒ c) order of rxn can not calculate, because it is an experimental quantity

☒ d) A & B both correct

Q-2 $2A \rightarrow B$ is an example of Complex Reaction

a) Molecularity = 2 $\rightarrow$ ~~x~~

b) order of rxn = 2 $\rightarrow$ ~~x~~

~~c~~ order of rxn can not calculate, because it is an experimental quantity

~~d~~ Molecularity of above reaction is not defined because molecularity only defines for elementary rxn

~~e~~ c & d both correct

Q-3. $2A + B \rightarrow p$, the rate law expression of above reaction is $\text{Rate} = k [A] [B]^{\frac{1}{2}}$. $\rightarrow 1 + \frac{1}{2} = \left(\frac{3}{2}\right)$

~~a)~~ order of rxn is 3

~~b)~~ order of rxn is $3/2$

~~c)~~ order of rxn is 1

~~d)~~ order of rxn is $\frac{1}{2}$

Q-4 $2A + 3B \rightarrow 5C$, the rate of disappearance of B is $-9.3 \times 10^2 \text{ mole} \cdot \text{litre}^{-1} \text{sec}^{-1}$. then Calculate Rate of reaction

a) $9.3 \times 10^2 \text{ mole} \cdot \text{litre}^{-1} \text{sec}^{-1}$

~~b) $3.1 \times 10^2 \text{ mole} \cdot \text{litre}^{-1} \text{sec}^{-1}$~~

c) $27.9 \times 10^2 \text{ mole} \cdot \text{litre}^{-1} \text{sec}^{-1}$

d) NONE

$$\begin{aligned} R.O.R. &= \frac{R_B}{3} = \frac{9.3 \times 10^2}{3} \\ &= 3.1 \times 10^2 \end{aligned}$$

Q-5 Select correct statements for order of rxn & molecularity of Reaction.

- a) Max. value of order of rxn & molecularity is 3
☒ T
- b) Value of order of rxn & molecularity may be fractional
☐ F ☒ T
- c) only order of rxn may be zero
☒ T
- d) Molecularity is Experimental value while order of rxn is theoretical
☐ F quantity.

a) a, d

~~b) a, c~~

c) a, c, d

d) a, b, d

THANK
.. you
😊